

Average number of hydrogen ionizations per primary particle: photons and electrons in the neutral IGM

Response note — Notebooks/redo/stage8_*

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Abstract

We compute $N_{\text{ion}}(E, z_i)$, the average total number of H I ionizations produced by one primary particle of energy E injected at redshift z_i into the fiducial neutral IGM, counting every ionization produced anywhere down the cascade until the end of the reionization window ($z = 5.5$). Two primary species are treated: electrons (existing calculation, `stage2_cascade.py`) and photons (new, `stage8_photon_yield.py`, photoionization + Compton scattering only, by request). The result is Fig. 1. Electrons obey the familiar $W \simeq 36.6(23)$ eV plateau between 100 eV and 30 keV; photons do not, because above ~ 2.3 keV the IGM becomes optically thin and most primaries leave the window without interacting.

1 Assumptions

Every assumption is stated where it enters; the ones that are global are collected here.

A1 — Background and gas. Flat Λ CDM with the *Planck* 2018 TT,TE,EE+lowE+lensing parameters (Planck Collaboration, 2020), as implemented in `astropy.cosmology.Planck18`. The neutral hydrogen density is the fiducial normalisation of the parent manuscript,

$$n_{\text{HI}}(z) = 1.8 \times 10^3 \text{ m}^{-3} \left(\frac{1+z}{21} \right)^3, \quad (1)$$

with a residual ionized fraction $x_e = 10^{-4}$ (free electrons enter only as Compton targets and as the f_{free} branch below). Helium, clumping, and any inhomogeneity are neglected: the medium is uniform at the mean density. The hydrogen ionization threshold is $B_{\text{H}} = 13.6057$ eV.

A2 — Window. Ionizations are counted only while $z > z_{\text{final}} = 5.5$; a particle that survives to z_{final} contributes nothing further. A second variant extends the count to $z > 3.0$ and is tabulated alongside (columns `Nt/Ypt`); it changes the photon yield by $\lesssim 6.5\%$ except in the transparency window near 10 keV, where it is +58% (Sect. 4).

A3 — Photon interaction channels. *By explicit instruction*, the photon cascade includes **photoionization of H(1s) and Compton scattering only**. Pair production on hydrogen nuclei (Bethe–Heitler) and on the CMB (Breit–Wheeler) are **omitted**. Both omitted channels can only *add* ionizations, so the photon curve is a **lower bound** wherever they matter; the two energies at which they switch on are computed in Sect. 3 and marked on Fig. 1.

A4 — Cross sections. Photoionization: the H(1s) cross section of Karzas & Latter (1961), normalised to $\sigma_{\text{pi}}(B_{\text{H}}) = 6.30 \times 10^{-22} \text{ m}^2$. Compton: the free-electron Klein–Nishina cross section (Klein & Nishina, 1929), both total and differential (Eq. 2). Regime of validity: the free-electron (impulse) treatment of scattering off a *bound* electron requires $T \gg B_{\text{H}}$; Sect. 3 D shows this is harmless here.

A5 — Bound versus free targets. Per hydrogen nucleus there are $1 + x_e$ electrons available for Compton scattering, a fraction $f_{\text{bound}} = 1/(1 + x_e)$ bound and $f_{\text{free}} = x_e/(1 + x_e)$ free. A Compton recoil T on a bound electron is itself one ionization and ejects an electron of kinetic energy $T - B_{\text{H}}$ when $T \geq B_{\text{H}}$, and ionizes nothing when $T < B_{\text{H}}$. On a free electron it ionizes nothing directly and simply hands over an electron of energy T .

A6 — Straight-line, light-speed photon transport. Photons travel on the null geodesic and redshift as $E \propto (1 + z)$; Compton deflection is not tracked (the medium is uniform, so only the *time* spent, not the direction, matters for the interaction budget).

A7 — Electron child yields. The number of ionizations produced by a secondary electron of kinetic energy K released at redshift z is taken from the existing table `redo_cascade_table.npz`, column $Y + Y_p^{(e)}$: the in-situ collisional cascade plus the ionizations made by the photons (bremsstrahlung, inverse Compton) that the electron itself radiates and that are re-absorbed inside the window. The underlying loss processes are inverse Compton and bremsstrahlung (Blumenthal & Gould, 1970), Coulomb losses (Gould, 1972), collisional excitation (Stone & Kim, 2002) and RBEB collisional ionization (Kim et al., 2000). Two inherited caveats:

- radiated photons are transported only up to $E_{\gamma}^{\text{max}} = 3 \times 10^4 \text{ eV}$ (above that the *photoionization* probability is $< 10^{-3}$ everywhere in the window). Compton absorption of those photons is not included in the electron table, so the electron curve is itself a mild lower bound above $K \sim 7 \times 10^8 \text{ eV}$, where inverse-Compton photons exceed $3 \times 10^4 \text{ eV}$. This is the origin of the electron plateau at $N_{\text{ion}} \simeq 4.1 \times 10^6$ in Fig. 1a.
- the child-yield lookup is clipped to $z \geq 7$, as in `stage2_cascade.py`: a sub-keV electron thermalises in $\Delta z \ll 1$, and the clipping costs 0.04%.

A8 — What is *not* assumed. No Monte Carlo, no continuous-slowng-down shortcut for the photon, and no iteration: the photon equation is solved exactly (Sect. 2).

2 Derivation

2.1 Klein–Nishina in the scattered-energy fraction

Write $a \equiv E/m_e c^2$ and $s \equiv E'/E$ for the fraction of the photon energy that survives the scattering. The Compton relation $E'/E = [1 + a(1 - \cos \theta)]^{-1}$ inverts to $\cos \theta = 1 - (1/s - 1)/a$ and gives $d \cos \theta = ds/(as^2)$. Substituting into the Klein–Nishina differential cross section (Klein & Nishina, 1929; Rybicki & Lightman, 1979, Eq. 7.5), $d\sigma/d \cos \theta = \pi r_e^2 s^2 (s + 1/s - 1 + \cos^2 \theta)$,

$$\frac{d\sigma}{ds} = \frac{\pi r_e^2}{a} \left[s + \frac{1}{s} - 1 + \cos^2 \theta \right], \quad \frac{1}{1 + 2a} \leq s \leq 1, \quad (2)$$

with recoil $T = E(1 - s)$, so that $E = E' + T$ identically. Equation (2) integrates back to the standard total cross section; this is verified symbolically in Sect. 3 A.

2.2 First-interaction weights along the light path

A photon injected at z_i with energy E has, at redshift $z < z_i$ along its path, energy $E(z) = E(1 + z)/(1 + z_i)$ and interaction rates

$$k_{\text{pi}}(z) = n_{\text{HI}}(z) \sigma_{\text{pi}}(E(z)) c, \quad k_{\text{C}}(z) = n_{\text{HI}}(z) (1 + x_e) \sigma_{\text{KN}}(E(z)) c. \quad (3)$$

With $\tau(t) = \int k_{\text{tot}} dt'$, the probability that the *first* interaction is of type j and occurs in dt is

$$dP_j = k_j(t) e^{-\tau(t)} dt. \quad (4)$$

The path is discretised in $\ln(1+z)$ on $N_{\text{path}} = 192$ nodes; on each segment τ is taken linear in t and $e^{-\tau}$ is integrated *analytically*, which makes every quadrature weight non-negative and guarantees $\sum_j \sum_{\text{path}} w_j = 1 - e^{-\tau_{\text{tot}}}$ exactly (Sect. 3 C). The branching between the two channels is $w_{\text{pi}} = w k_{\text{pi}}/k_{\text{tot}}$, $w_{\text{C}} = w k_{\text{C}}/k_{\text{tot}}$.

2.3 The cascade equation and how it is solved

Let $N(E, z_i)$ be the quantity asked for. Collecting Eqs. (2)–(4) and A5,

$$\begin{aligned} N(E, z_i) = & \sum_{\text{path}} w_{\text{pi}} \left[1 + Y_e(E(z) - B_{\text{H}}, z) \right] \\ & + \sum_{\text{path}} w_{\text{C}} \int_{1/(1+2a)}^1 ds p(s | E(z)) \left\{ f_{\text{bound}} \Theta(T - B_{\text{H}}) [1 + Y_e(T - B_{\text{H}}, z)] \right. \\ & \left. + f_{\text{free}} Y_e(T, z) + N(E(z) s, z) \right\}, \end{aligned} \quad (5)$$

where $p(s|E)$ is Eq. (2) normalised to unity, $T = E(z)(1-s)$, and Y_e is the electron child yield of A7. The three terms on the second and third lines are, in order: ionization of a bound target plus the cascade of the ejected electron; the cascade of a Compton-heated free electron; and the *continuation of the scattered photon*, which is what makes Eq. (5) an integral equation.

Equation (5) is a Volterra equation in E : every argument on the right-hand side is $\leq E$, with equality only in the joint limit $z \rightarrow z_i$, $s \rightarrow 1$. It is therefore solved by marching *upward* in energy on a logarithmic grid of 120 nodes spanning $13.6\text{--}1 \times 10^{13}$ eV, exactly as `stage2_cascade.py` marches upward in K . The near-elastic Compton limit ($E \ll m_e c^2$, where the whole scattered-photon weight falls inside one grid cell) is handled without iteration: the bilinear interpolation weight that lands on the current energy node is kept on the left-hand side, so each energy node reduces to one exact linear system over the $N_z = 22$ redshift rows,

$$(I - M) \vec{N}_{:,i} = \vec{A}, \quad (6)$$

solved by `numpy.linalg.solve`. No relaxation, no convergence criterion: the multiple-scattering chain is resummed exactly. The matrix is extremely well conditioned because the *absolute* first-interaction Compton weight never exceeds 0.151 anywhere on the grid (it is bounded by the total interaction probability, not by the branching ratio).

The s -integral uses a 64-node Gauss–Legendre rule in $u = \ln s$: the integrand of Eq. (2) has a $1/s$ tail spanning up to seven decades at $E = 5$ TeV, which is flat in u , while for $a \ll 1$ the interval shrinks as $2a$ and the rule remains exact.

3 Verification and cross-checks

Full output: `Notebooks/redo/stage8_verify.log`. All checks pass.

A — Klein–Nishina, differential $\rightarrow$ total. (i) *Symbolic.* `sympy` integrates Eq. (2) over $s \in [1/(1+2a), 1]$ for symbolic $a > 0$; the difference from the closed-form total cross section `simplifys` to exactly 0, with numeric residuals $\leq 2 \times 10^{-124}$ at $a = 10^{-3}, 0.1, 1, 10, 10^3$. (ii) *Independent quadrature.* `mpmath` adaptive quadrature at 30 digits reproduces the closed form to 1.2×10^{-16} relative at every energy from 1 eV to 1×10^{13} eV. (iii) *The rule actually used.* The 64-node Gauss–Legendre rule in $\ln s$ agrees with the closed form to 4.4×10^{-7} relative, worst case, over $13.6\text{--}1 \times 10^{13}$ eV. (iv) *Thomson limit.* The quadrature reproduces the low-energy expansion $\sigma_{\text{KN}}/\sigma_{\text{T}} = 1 - 2a + \frac{26}{5}a^2 - \frac{133}{10}a^3$ (Heitler, 1954, §22) to all displayed digits. *Flagged:* the float64 evaluation of the closed-form σ_{KN} in `redo_common.py` loses 4×10^{-7} relative accuracy at 10 eV and 10^{-5} at 1 eV to cancellation. This is far below every other error here and is recorded, not fixed.

B — Compton kinematics. All quadrature nodes lie inside $[1/(1+2a), 1]$; the analytic endpoint $s_{\min} = 1/(1+2a)$ and $T_{\max} = E 2a/(1+2a)$ are reproduced; $E = E' + T$ holds identically by construction.

C — Path kernel. At every (z_i, E) tested, all weights are non-negative and $\sum w = 1 - e^{-\tau_{\text{tot}}}$ to better than 0.5% against an independent 4×-finer trapezoidal evaluation of τ_{tot} (typically to 10^{-5} relative).

D — Channel competition and the impulse approximation. $\sigma_{\text{pi}} = \sigma_{\text{C}}$ at $E = 2.333 \text{ keV}$. The maximum Compton recoil $T_{\max} = E 2a/(1+2a)$ reaches B_{H} only at $E = 1.871 \text{ keV}$. Below that energy *every* Compton event is sub-threshold: the model produces no ionization and returns the photon with $E' \simeq E$, which is what bound-electron (Rayleigh/Raman) scattering does as well, so the free-electron treatment cannot bias the ionization budget there. Above it, $T \gg B_{\text{H}}$ within a factor of a few and the impulse approximation is the standard one (Zdziarski & Svensson, 1989). Photoionization dominates the total interaction rate all the way to 2.3 keV, so Compton never controls the yield below $\sim 2 \text{ keV}$.

E — Omitted channel I: Bethe–Heitler pair production on H. Using the high-energy no-screening formula $\sigma = 4\alpha r_e^2 Z(Z+1)[\frac{7}{9} \ln 2k - \frac{109}{54}]$ and the complete-screening formula $\sigma = 4\alpha r_e^2 Z(Z+1)[\frac{7}{9} \ln(183Z^{-1/3}) - \frac{1}{54}]$ (Heitler, 1954; Motz et al., 1969, §26), with $Z = 1$ and $Z \rightarrow Z(Z+1) = 2$ for triplet production, the complete-screening asymptote is $\sigma_{\text{pair}} = 1.87 \times 10^{-30} \text{ m}^2$ and $\sigma_{\text{pair}} = \sigma_{\text{KN}}$ at

$$E_{\text{pair}} = 54 \pm 17 \text{ MeV} \quad (\text{the two screening limits bracket the true crossing}). \quad (7)$$

Above the conservative edge 37 MeV the photon curve is a lower bound.

F — Omitted channel II: pair production on the CMB. Integrating the Breit–Wheeler cross section (Breit & Wheeler, 1934)(Jauch & Rohrlich, 1976, §11-3) over the full Planck spectrum at $T_{\text{CMB}} = 2.7255(1+z) \text{ K}$, and requiring the mean free path (evaluated at the injection redshift, where the CMB is hottest and the onset earliest) to fall below the proper light path down to $z = 5.5$, gives

z_i	light path [m]	$E_{\gamma\gamma} \text{ [eV]}$
20	8.14×10^{24}	2.44×10^{12}
15	7.29×10^{24}	3.33×10^{12}
10	5.36×10^{24}	5.25×10^{12}
7	2.63×10^{24}	7.87×10^{12}

so the top decade of the grid is also a lower bound. (The naive threshold against a typical CMB photon, $(m_e c^2)^2/kT$, is 53 TeV at $z = 20$; the Wien tail brings the onset down by a factor ~ 20 .)

G — Numerical convergence. The complete march was repeated with every discretisation doubled: $N_{\text{path}}: 192 \rightarrow 384$, $N_s: 64 \rightarrow 128$, energy nodes $120 \rightarrow 239$. Both tables are interpolated log–log to the same energies (comparing nearest grid nodes would report the grid offset, not the error). The largest change anywhere in $7 \leq z_i \leq 20$, $20\text{--}1 \times 10^{12} \text{ eV}$ is **1.7%**, at $E = 10 \text{ keV}$, $z_i = 10$ — the transparency minimum, where the curve has its sharpest curvature in $\ln E$ and the interpolation itself dominates the difference. Outside the 0.5–10 keV transparency band the change is below **0.5%** everywhere, and below 2×10^{-5} above 1 GeV. **We therefore quote a numerical uncertainty of 0.5% on the tabulated yields, rising to 2% in the 0.5–10 keV band.**

H — Physical limits (independent check of the full march). Where the IGM is optically thick the answer is known in closed form: the photon is certainly photoionized, giving one ionization plus the cascade of a photoelectron of energy $E - B_H$, i.e. $N_\gamma(E) = 1 + Y_e(E - B_H)$ exactly. The march, which knows nothing of this shortcut, reproduces it:

E [eV]	20.4	90.6	297	478	975
$N_\gamma/[1 + Y_e(E - B_H)], z_i = 20$	1.0000	0.9999	0.9968	0.9835	0.878
$N_\gamma/[1 + Y_e(E - B_H)], z_i = 10$	1.0000	0.9998	0.9916	0.9602	0.688

The agreement is exact to four digits wherever $P_{\text{int}} = 1$, and departs only as the photon starts to survive to z_{final} ($P_{\text{int}} = 0.856$ at 975 eV for $z_i = 10$), which is the physically correct behaviour. At threshold the table gives $N = 0.99989$ at 13.61 eV and $N = 1.00000$ at 20.4 eV, as it must.

4 Final result

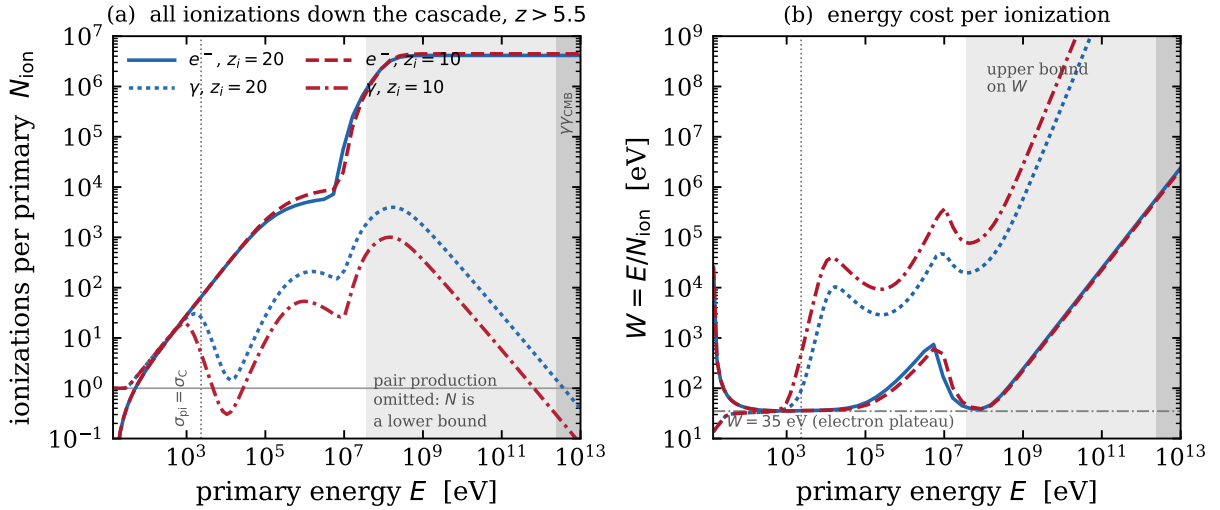


Figure 1: Average number of hydrogen ionizations produced by one primary particle of energy E injected at z_i , counting all ionizations down the cascade until $z = 5.5$. Solid/dashed: electrons ($z_i = 20, z_i = 10$); dotted/dash-dotted: photons. Vertical dotted line: $\sigma_{\text{pi}} = \sigma_{\text{C}}$ at 2.33 keV. Light shading: pair production on H omitted, so N is a lower bound ($E > 37$ MeV). Dark shading: pair production on the CMB also omitted ($E > 2.4$ TeV at $z_i = 20$). (b) The same as the energy cost per ionization $W = E/N_{\text{ion}}$; the dash-dotted grey line is the canonical 35 eV.

Reading of the figure

Electrons. $W = K/N_{\text{ion}}$ is flat at 36.6(23) eV (min 35.38, max 40.03, mean 36.63 over $100 - 3 \times 10^4$ eV at $z_i = 20$; 36.55 eV at $z_i = 10$), the standard result for a cascade in a nearly neutral medium (Shull & van Steenberg, 1985; Furlanetto & Stoever, 2010). Above ~ 30 keV the electron range starts to exceed the window and W climbs; the sharp rise of N_{ion} between $1 \times 10^7 - 1 \times 10^9$ eV is the onset of inverse-Compton emission into the absorbable band ($E_\gamma \lesssim 3 \times 10^4$ eV), and the plateau at 4.1×10^6 above 1×10^9 eV is the point where the IC photons leave that band (A7).

Photons. Three regimes:

- 13.6–500 eV: the IGM is optically thick to photoionization, $P_{\text{int}} = 1$, and the photon behaves *exactly* like one ionization plus an electron of energy $E - B_H$ (Sect. 3H). Because of that

Table 1: N_{ion} per primary particle. Numerical uncertainty is the convergence bound of Sect. 3 G; the systematic uncertainty from the omitted photon channels is one-sided (the true value is larger) and is indicated by the shading in Fig. 1.

E [eV]	electron		photon	
	$z_i = 20$	$z_i = 10$	$z_i = 20$	$z_i = 10$
10^2	2.66	2.66	2.81	2.81
10^3	28.3	28.3	24.8	19.5
10^4	323	326	1.65	0.303
10^5	2.09×10^3	2.36×10^3	22.8	7.24
10^6	4.88×10^3	6.76×10^3	198	53.7
10^7	5.53×10^4	1.94×10^4	226	27.8
10^8	2.24×10^6	2.22×10^6	3.75×10^3	981
10^{10}	4.14×10^6	4.46×10^6	231	46.7
10^{12}	4.14×10^6	4.46×10^6	3.63	0.726

extra ionization the photon curve sits slightly *above* the electron curve at the same energy (by 27% at 90 eV, 7% at 300 eV), the two merging as E grows. Between 500–2300 eV the photon starts to survive the window and the curves separate downward.

2. 2.3–100 keV: the transparency window. $\sigma_{\text{pi}} \propto E^{-3}$ collapses, and the total interaction probability at $z_i = 20$ falls from 1 at 2 keV to 0.157 at 10 keV. N_{ion} has a sharp minimum, $N = 1.45$ at 13.3 keV ($z_i = 20$) and $N = 0.303$ at 10.5 keV ($z_i = 10$): *a 10 keV photon injected at $z = 10$ produces, on average, less than one ionization before the end of reionization*. This is the single most important feature of the figure and the qualitative difference from the electron case.
3. > 100 keV: Compton takes over. The interaction probability keeps falling (6.4×10^{-2} at 1 MeV, 2.8×10^{-3} at 100 MeV, 5.4×10^{-5} at 10 GeV for $z_i = 20$) but the yield *per* interaction grows faster, because the Compton recoil approaches E and the resulting electron is itself an efficient ioniser. The product peaks at $N \simeq 3.9 \times 10^3$ near 1.4×10^8 eV and then declines as $N \propto E^{-1}$ once the electron yield saturates.

Window sensitivity (A2). Extending the count from $z > 5.5$ to $z > 3.0$ changes the photon yield at $z_i = 20$ by +0.0% (1 keV), +1.9% (89 keV), +6.5% (1 MeV) and +4.9% (100 MeV) — but by +58% at 10.5 keV, precisely in the transparency window, where the result is dominated by how long the photon is allowed to keep travelling. Quoted photon yields near 10 keV are therefore window-definition dependent at the tens-of-per-cent level; everywhere else they are not.

5 Why UV photons, not energetic particles, reionize the Universe

Figure 1a is easily misread as saying that electrons are the more effective ionizing agents, since a 1 GeV electron makes 4×10^6 ionizations and no photon on the grid makes more than 4×10^3 . That reading confuses two different questions. The figure is *per primary particle*; reionization is decided by

$$\mathcal{N}_{\text{ion}} = \underbrace{\frac{1}{W(E)}}_{\text{ionizations per erg}} \times \underbrace{\mathcal{E}_{\text{channel}}}_{\text{ergs available}}, \quad (8)$$

and *both* factors favour the ultraviolet. Panel (b) of Fig. 1 already contains the first factor; the second is source physics and is taken from the literature. The combination is Fig. 2, computed by `stage9_budget.py`.

5.1 Assumptions of this section

The inputs are listed with their provenance; none is invented.

symbol	value	meaning	source
ξ_{ion}	$10^{25.2} \text{ Hz erg}^{-1}$	ionizing photon prod. eff.	Robertson et al. (2015); Bouwens et al. (2016)
$L_{\nu}^{\text{UV}}/\text{SFR}$	$(1.15 \times 10^{-28})^{-1} \text{ erg s}^{-1} \text{ Hz}^{-1} \text{ per } M_{\odot} \text{ yr}^{-1}$	UV luminosity per SFR	Kennicutt (1998); Madau & Dickinson (2014)
$\langle E_{\gamma} \rangle$	20 eV	mean ionizing photon energy	Pop II population, 18–25 eV
f_{esc}	0.1	UV escape fraction	Robertson et al. (2015); Finkelstein et al. (2019)
L_X/SFR	$3 \times 10^{39-40} \text{ erg s}^{-1} \text{ per } M_{\odot} \text{ yr}^{-1}$	HMXB X-rays, 0.5–8 keV	Mineo et al. (2012); Fragos et al. (2013); Lehmer et al. (2021)
Γ_X	2	HMXB photon index	Frags et al. (2013)
$\mathcal{E}_{\text{CR}}/M_{\star}$	$1.4 \times 10^{48} \text{ erg } M_{\odot}^{-1}$	all cosmic rays, <i>measured</i>	Strong et al. (2010) / Licquia & Newman (2015)
$\mathcal{E}_{\text{CRE}}/M_{\star}$	$2.1 \times 10^{46} \text{ erg } M_{\odot}^{-1}$ (range $1.5\text{--}2.9 \times 10^{46}$)	CR <i>electrons</i> , <i>measured</i>	Strong et al. (2010) / Licquia & Newman (2015)
K_{ep}	0.012–0.021 (measured)	CR electron/proton injected-power ratio	Strong et al. (2010), Table 2
<i>cross-check chain for the two CR rows, recomputed in <code>stage9_verify_budget.py</code></i>			
E_{SN}	$10^{51} \text{ erg per CCSN}$	canonical explosion energy	Woosley & Weaver (1995)
$M_{\star}/N_{\text{CCSN}}$	84.8 $M_{\odot}$ (Chabrier) 135 $M_{\odot}$ (Salpeter)	recomputed from the IMF here, 8–100 $M_{\odot}$ progenitors	Chabrier (2003); Salpeter (1955)
ε_{CR}	0.1 (range 0.1–0.5)	SN shock kinetic energy $\rightarrow$ cosmic rays	Drury et al. (1989); Caprioli & Spitkovsky (2014), via Gessey-Jones et al. (2023)

Uncertainties on these are factors of a few, not orders of magnitude, except L_X/SFR and ε_{CR} , which are scanned explicitly below. Everything is normalised per solar mass of star formation, so the comparison is independent of the star formation history.

5.2 First factor: the optimum is at threshold, not at high energy

Table 2 reads $1/W$ off the stage 8 tables at $z_i = 10$.

Table 2: Ionizations per erg of injected energy, from Fig. 1b. The last column is normalised to a 20 eV UV photon.

channel	W [eV]	ionizations erg^{-1}	relative
UV photon, 20 eV	20.4	3.06×10^{10}	1.000
UV photon, 30 eV	28.5	2.19×10^{10}	0.717
soft X-ray, 0.5 keV	35.7	1.75×10^{10}	0.572
X-ray, 1 keV	52.5	1.19×10^{10}	0.389
X-ray, 3 keV	1.08×10^3	5.80×10^8	0.019
hard X-ray, 10 keV	3.21×10^4	1.95×10^7	6.4×10^{-4}
electron, 1 keV	35.4	1.77×10^{10}	0.577
electron, 1 MeV	149	4.20×10^9	0.137
CR electron, 1 GeV	225	2.77×10^9	0.091
CR electron, 1 TeV	2.24×10^5	2.78×10^6	9.1×10^{-5}
HMXB power law, $\Gamma = 2$, 0.5–8 keV	110	5.68×10^9	0.186

Three statements follow, all of them visible in Fig. 1b and none of them obvious from Fig. 1a.

(i) **A threshold photon is the perfect ioniser.** A photon of 13.6 eV deposits *all* of its energy in one ionization, $W = B_H$, which is the thermodynamic optimum: no agent can do better. Our own table gives $N = 1.00000$ at 20.4 eV. A fast electron cannot reach it, because it spends roughly a third of its energy on collisional excitation ($\text{Ly}\alpha$) and a third on Coulomb heating for every third spent on ionization (Shull & van Steenberg, 1985; Valdes & Ferrara, 2008; Furlanetto & Stoevers, 2010); hence the familiar $W \simeq 35$ eV floor rather than 13.6 eV. *Per erg*, UV photons beat the best possible electron by a factor $\simeq 2$.

(ii) **Above 2 keV the IGM stops absorbing.** This is the transparency window found in Sect. 4: the bound-free cross section falls as $\sigma_{\text{pi}} \propto E^{-3}$, the crossover to Compton occurs at 2.33 keV (Sect. 3 D), and the total interaction probability inside the window collapses from 1 at 2 keV to 0.157 at 10 keV ($z_i = 20$). This is the standard statement that hard X-rays have a mean free path comparable to the Hubble scale (Oh, 2001; Venkatesan et al., 2001; McQuinn, 2012; Mesinger et al., 2013): they do not fail to ionize *efficiently*, they fail to interact *at all* within the window. The consequence in Table 2 is a factor 1.6×10^3 penalty between 0.5 keV and 10 keV.

(iii) **High-energy electrons are also inefficient per erg.** A 1 GeV electron makes 4×10^6 ionizations, but it costs 1 GeV to make: $W = 225$ eV, an order of magnitude worse than UV. The reason is visible in Fig. 1b: above ~ 1 MeV the electron cools predominantly by inverse Compton scattering on the CMB, and most of the up-scattered photons land above 3×10^4 eV, i.e. inside the transparency window of (ii), and leave. So the two failure modes are the same failure mode.

5.3 Second factor: the energy budgets differ by three to four orders of magnitude

Table 3: Energy available per solar mass of star formation, and the ionizations it delivers when multiplied by Table 2. Note the deliberate asymmetry in f_{esc} (see text).

channel	erg $M_{\odot}^{-1}$	$N_{\text{ion}} M_{\odot}^{-1}$	relative to UV
stellar H-ionizing UV (intrinsic)	1.39×10^{50}	—	—
escaping, $f_{\text{esc}} = 0.1$	1.39×10^{49}	4.26×10^{59}	1
HMXB X-rays, $L_X/\text{SFR} = 3 \times 10^{39}$	9.5×10^{46}	5.4×10^{56}	1.3×10^{-3}
HMXB X-rays, $L_X/\text{SFR} = 10^{40}$	3.2×10^{47}	1.8×10^{57}	4.2×10^{-3}
cosmic rays, all species	1.44×10^{48}	$\sim 4.0 \times 10^{57\dagger}$	$\sim 9.4 \times 10^{-3\dagger}$
cosmic-ray <i>electrons</i>	2.1×10^{46}	5.7×10^{55}	1.3×10^{-4}
range from the MW anchor	$1.5\text{--}2.9 \times 10^{46}$	—	$(1.0\text{--}1.9) \times 10^{-4}$

[†] **Low confidence, flagged.** No proton cascade was computed in this project. The entry assumes a proton W comparable to the electron one and is indicative only; the published limit is “no more than a few per cent”

(Sazonov & Sunyaev, 2015; Leite et al., 2017), with which it agrees.

The intrinsic UV budget is 1.4×10^{50} erg $M_{\odot}^{-1}$ (4.3×10^{60} ionizing photons per $M_{\odot}$, i.e. $\log_{10} N_{\text{ion}}/M_{\odot} = 60.64$, the standard value). Even after the $f_{\text{esc}} = 0.1$ penalty it exceeds the HMXB X-ray budget by $\sim 40\text{--}150$ and the cosmic-ray *electron* budget by 1.4×10^3 .

Two points deserve emphasis because they are the ones usually left implicit.

The escape asymmetry is real and it is the hard channels’ only advantage. The UV number carries $f_{\text{esc}} \simeq 0.1$ while the X-ray and cosmic-ray numbers carry $f_{\text{esc}} \simeq 1$: the same E^{-3} cross section that makes the IGM transparent to X-rays makes the *host galaxy* transparent to them as well. That asymmetry is worth a factor ~ 10 to the hard channels and is precisely why they are discussed at all (Oh, 2001; McQuinn, 2012). It is still not nearly enough.

Where the cosmic-ray electron budget comes from ($2.1 \times 10^{46} \text{ erg } M_{\odot}^{-1}$). This number is *measured*, not assembled from assumptions. Strong et al. (2010) fit the GALPROP propagation model to direct cosmic-ray measurements, non-thermal radio, hard X-ray and γ -ray data, and report in their Table 2 the *injected* cosmic-ray luminosity of the Milky Way split by species over six propagation models: $6.6\text{--}8.1 \times 10^{40} \text{ erg s}^{-1}$ in total (headline value 7.9×10^{40}), $6.0\text{--}7.4 \times 10^{40}$ in protons and $8.65 \times 10^{38}\text{--}1.34 \times 10^{39} \text{ erg s}^{-1}$ in *primary electrons*. Dividing by the Milky Way star formation rate, $\text{SFR} = 1.65 \pm 0.19 M_{\odot} \text{ yr}^{-1}$ (Licquia & Newman, 2015), gives

$$\frac{\mathcal{E}_{\text{CRE}}}{M_*} = \frac{L_{e^-}^{\text{prim}}}{\text{SFR}} = (1.5\text{--}2.9) \times 10^{46} \text{ erg } M_{\odot}^{-1}, \quad \frac{\mathcal{E}_{\text{CR}}}{M_*} = 1.4 \times 10^{48} \text{ erg } M_{\odot}^{-1}, \quad (9)$$

the quoted range spanning the six propagation models and the SFR error. The same table also *measures* the electron-to-proton injected-power ratio, $K_{ep} = L_{e^-}^{\text{prim}}/L_p = 0.012\text{--}0.021$.

The conventional assumption chain — $\varepsilon_{\text{CR}} E_{\text{SN}} N_{\text{CCSN}}/M_*$ with $\varepsilon_{\text{CR}} = 0.1$, $E_{\text{SN}} = 10^{51} \text{ erg}$ and one core-collapse supernova per $\sim 100 M_{\odot}$ — is an independent route to the same quantity, and `stage9_verify_budget.py` recomputes its IMF factor from scratch: integrating the Chabrier (2003) disc IMF over $8\text{--}100 M_{\odot}$ progenitors gives $84.8 M_{\odot}$ per core-collapse supernova (Salpeter 1955: $135 M_{\odot}$), i.e. $E_{\text{SN}}/M_* = 1.2 \times 10^{49} \text{ erg } M_{\odot}^{-1}$ and $\mathcal{E}_{\text{CRE}}/M_* \simeq 1.2 \times 10^{46}$. The chain and the measurement agree to a factor 1.7, which is the honest uncertainty on this row.

Cosmic-ray *electrons* are the wrong tracer of the cosmic-ray budget. Equation (9) makes the point directly: electrons carry $\sim 1.5\%$ of the injected cosmic-ray power, the rest being protons and helium (Strong et al., 2010; Gessey-Jones et al., 2023). The cosmic-ray reionization literature (Sazonov & Sunyaev, 2015; Leite et al., 2017) is therefore about *protons*, whose budget is $\sim 70\times$ larger than the electron budget plotted in Fig. 1. Our electron curve, striking as it looks per particle, has essentially no counterpart in a realistic source budget.

Caveat on transplanting the Milky Way to $z \sim 10$. The anchor is a present-day, high-metallicity, strongly magnetised disc. At $z \sim 10$ both the supernova energy per $M_{\odot}$ (a top-heavier IMF, more massive progenitors) and the fraction of cosmic rays escaping into the IGM (smaller haloes, weaker fields; Leite et al. 2017) are plausibly larger. Both push the same way, and neither is a factor 10^4 : a full order-of-magnitude enhancement would still leave cosmic-ray electrons at $\sim 10^{-3}$ of the UV ionizations. Adopting instead the $K_{ep} \simeq 0.1$ sometimes used for extragalactic sources (Merten et al., 2017) changes the same conclusion by one order of magnitude and no more.

5.4 The result, and the comparison with the literature

Multiplying the two factors (Fig. 2b):

$$\frac{\mathcal{N}_{\text{ion}}^{\text{X}}}{\mathcal{N}_{\text{ion}}^{\text{UV}}} \simeq 1\text{--}4 \times 10^{-3}, \quad \frac{\mathcal{N}_{\text{ion}}^{\text{CR } e^-}}{\mathcal{N}_{\text{ion}}^{\text{UV}}} \simeq 1.3 \times 10^{-4}, \quad \frac{\mathcal{N}_{\text{ion}}^{\text{CR } p}}{\mathcal{N}_{\text{ion}}^{\text{UV}}} \sim 9 \times 10^{-3} \uparrow. \quad (10)$$

Scanning the X-ray SED over $\Gamma_X \in [1.5, 2.5]$ and a low-energy escape cutoff $E_{\text{min}} \in [0.2, 1] \text{ keV}$ at the optimistic $L_X/\text{SFR} = 10^{40}$ gives 1.2×10^{-3} to 9.3×10^{-3} : even the most favourable corner (softest spectrum, lowest cutoff) stays at the per-cent level. Reaching the $\sim 10\%$ X-ray share that has been argued for in the literature would require $f_{\text{esc}}(\text{UV}) \sim 0.01$ or $L_X/\text{SFR} \sim 10^{42}$; the latter is exactly what the unresolved soft X-ray background excludes (McQuinn, 2012), and Mesinger et al. (2013) find that X-ray feedback can sustain a $\sim 10\%$ ionized IGM but cannot complete reionization. Our numbers therefore sit at the conservative end of, and are consistent with, the published range.

A fourth argument that is not about efficiency at all. Hard channels are self-limiting. As x_e rises, a smaller fraction of the energy of a fast electron goes into further ionization and a larger fraction into heat (Shull & van Steenberg, 1985; Valdes & Ferrara, 2008; Furlanetto & Stoeve, 2010; Chen & Kamionkowski, 2004). The present project’s own stage 5 table (see the “Lock coefficient” block earlier in RESULTS.txt) quantifies this with the same code base: $W(1\text{ keV})$ rises from 35.3–89.7 eV as x_e goes from 10^{-4} to 0.1. A UV photon above 13.6 eV suffers no such penalty: it is absorbed by a neutral atom or it is not absorbed at all. X-rays and cosmic rays can therefore drive the IGM to a partially ionized, pre-heated state — which is their accepted role, and the one relevant to the 21-cm signal (Mesinger et al., 2013) — but they cannot take x_e to unity.

And a fifth, from the observed history. The UV budget only just closes. Requiring $1 + N_{\text{rec}} \simeq 2$ ionizing photons per hydrogen atom with $f_{\text{esc}} = 0.1$ needs $\rho_* \simeq 2.6 \times 10^7 M_\odot \text{ Mpc}^{-3}$ comoving, against an observed $z \sim 6$ stellar mass density of order $10^7 M_\odot \text{ Mpc}^{-3}$ (Madau & Dickinson, 2014). This is the well-known photon-starved character of reionization (Bolton & Haehnelt, 2007). Reionization also ends rapidly, by $z \simeq 5.3$ (Bosman et al., 2022), with $\tau_e = 0.054$ (Planck Collaboration, 2020) — a sharp, ionization-front-driven transition, not the gradual, spatially smooth, partially-ionized history that a hard-photon or cosmic-ray-dominated model produces (Mesinger et al., 2013). Both the budget and the morphology point the same way.

5.5 So: is Fig. 1 consistent with the standard picture?

Yes, and it quantifies exactly why. There is no inconsistency, only a change of variable. Specifically:

1. Read *per erg* rather than per particle — panel (b), not panel (a) — the figure says the optimum ioniser is a photon at the hydrogen threshold and that everything else is worse, monotonically in the sense that matters. Nothing in the figure ever goes above the horizontal “one ionization per B_H ” line in Fig. 2a, because nothing can.
2. The apparent superiority of electrons at fixed E above $\sim 1\text{ keV}$ is *transport*, not intrinsic efficiency: a keV electron stops locally (range $\ll$ window) while a keV photon has a mean free path approaching the Hubble scale. Below 500 eV the ordering actually reverses and the *photon* is the better ioniser, by exactly one extra ionization ($N_\gamma = 1 + Y_e(E - B_H)$, Sect. 3H) — +27% at 90 eV.
3. The transparency minimum at 3–30 keV, where a photon injected at $z = 10$ makes *less than one* ionization, is the quantitative form of the textbook statement that the IGM is transparent to hard X-rays. It is the single most reionization-relevant feature in the figure.
4. Where the figure genuinely adds something to the standard picture is the shape of the high-energy photon curve: because the omitted pair-production channels are lower bounds (Sect. 3E,F), the true N_{ion} above 37 MeV is larger than plotted — but W there is $\gtrsim 2.7 \times 10^4\text{ eV}$, so no plausible correction can make that band competitive.

Confidence flags for this section. The $1/W$ column of Table 2 inherits the 0.5% numerical uncertainty of Sect. 3G and, above 37 MeV, the one-sided systematic of the omitted channels. The energy budgets in Table 3 carry factor-of-a-few uncertainties, dominated by f_{esc} , $\langle E_\gamma \rangle$ and L_X/SFR ; the cosmic-ray rows are the best constrained of the three, since they are measured (Eq. 9) rather than assumed; *these are the reason no result here is quoted to better than one significant figure*. The cosmic-ray proton entry is not a calculation and is flagged as such. The conclusion is robust to all of them, because it rests on a margin of 10^2 – 10^4 , not on the precision of any single input.

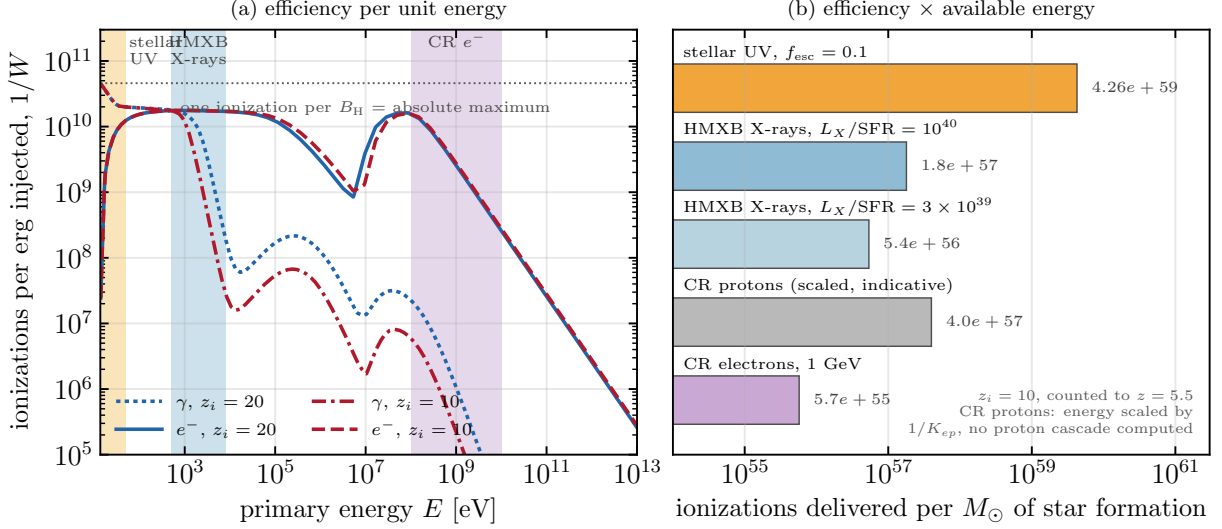


Figure 2: (a) Ionizations per erg of injected energy, $1/W(E)$, the quantity that actually matters for reionization, with the characteristic energies of the three source classes shaded. The dotted line is the absolute maximum, one ionization per B_H ; a threshold UV photon sits within a factor 1.5 of it and nothing else comes close. (b) The same efficiency multiplied by the energy budget of each channel, per solar mass of star formation (Table 3). The cosmic-ray proton bar is indicative only (no proton cascade was computed; see the flag on Table 3).

Reproducibility

- Notebooks/redo/stage8_photon_yield.py — builds redo_photon_yield_table.npz (181s). Parameters: $N_{\text{path}} = 192$, $N_s = 64$, 120 energy nodes ($13.61\text{--}1 \times 10^{13}$ eV), 22 redshift rows ($3.0 \leq z \leq 20.0$), $x_e = 10^{-4}$, $z_{\text{final}} = 5.5$ (Ne) and 3.0 (Nt).
- Notebooks/redo/stage8_verify.py → stage8_verify.log — all checks of Sect. 3.
- Notebooks/redo/stage8_converge.py → stage8_converge.log — the doubled-resolution run.
- Notebooks/redo/stage8_figure.py → manuscript/figures_redo/fig_ionizations_per_primary.{pdf,png}.
- Electron input: redo_cascade_table.npz from stage2_cascade.py (unchanged).
- Notebooks/redo/stage9_budget.py → stage9_budget.log — Sect. 5, all budget numbers.
- Notebooks/redo/stage9_verify_budget.py → stage9_verify_budget.log — provenance of the cosmic-ray budget: the Milky Way anchor, the IMF factor recomputed from Chabrier (2003) and Salpeter (1955), and the agreement between the two routes.
- Notebooks/redo/stage9_figure.py → manuscript/figures_redo/fig_reionization_budget.{pdf,png}.
- No stochastic method is used anywhere, so no random seed is involved.

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